

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 30%

QUESTIONS: 2 Total Marks:30 Annexure A

Industry Problem Solving Task

Code of Conduct:

I VALLAPU MOHAN KUMAR (192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Mohan Kumar

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Industry Problem	20%	Comprehensive understanding of real world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	

Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Annexure A

Industry Problem Solving Task

1: Automated Defect Detection in Manufacturing

A precision engineering company wants to automatically detect defects in mechanical components using computer vision. Parts vary slightly in shape due to manufacturing tolerances.

Questions:

1. How can PCA-based shape priors and contour-based representations be used to model standard component shapes?

Answer:

- Collect contour images of several defect-free components.
- Extract the object boundary using edge detection and contour extraction.
- Represent each contour as a set of ordered boundary points.
- Apply Principal Component Analysis (PCA) to learn the common shape variations.
- PCA creates a statistical model of the standard component shape.
- During inspection, compare the test component with the PCA model.
- If the deviation exceeds a threshold, classify it as a defective component.

2. Propose a pipeline using machine learning (SVM, GMM) or deep learning (CNN) to classify defective vs. non-defective parts.

Answer:

Pipeline:

1. Image Acquisition
2. Image Preprocessing
 - Noise removal
 - Contrast enhancement

3. Segmentation
4. Feature Extraction
 - Shape
 - Texture
 - Edge features
5. Feature Selection
6. Classification
 - SVM for binary classification
 - GMM for probabilistic clustering
 - CNN for automatic feature learning
7. Defect Detection
8. Performance Evaluation using Accuracy, Precision, Recall and F1-score.

3. Discuss methods to handle noise, partial occlusions, and varying illumination in industrial images.

Answer:

Noise Handling

- Gaussian Filter
- Median Filter
- Bilateral Filter

Partial Occlusion Handling

- Data augmentation
- CNN with attention mechanism
- Robust contour matching

Illumination Handling

- Histogram Equalization
- CLAHE

- Adaptive Thresholding
- Image Normalization

These methods improve defect detection accuracy under challenging industrial conditions.

2: Autonomous Vehicle Obstacle Recognition

A self-driving car company needs to detect rare or unexpected obstacles on urban roads. The labeled data for these obstacles is limited.

Questions:

1. Explain how data augmentation using GANs can help generate synthetic images for training.

Answer:

Generative Adversarial Networks (GANs) are deep learning models used to create realistic synthetic images that resemble real-world images. A GAN consists of two neural networks:

- **Generator:** Generates fake (synthetic) images from random noise.
- **Discriminator:** Determines whether an image is real or generated.

During training, the Generator continuously improves by trying to fool the Discriminator, while the Discriminator learns to distinguish real images from fake ones. This adversarial process enables the Generator to produce highly realistic images.

For obstacle recognition in autonomous vehicles, GANs can generate synthetic images of **rare or unexpected obstacles** (such as fallen trees, animals, debris, or damaged vehicles), which are difficult to collect in large numbers. These synthetic images are added to the original training dataset as **data augmentation**.

Benefits of GAN-based data augmentation:

- Increases the size and diversity of the training dataset.
- Generates realistic images of rare obstacles.
- Reduces overfitting by exposing the model to more variations.
- Improves the robustness and generalization of CNN or Vision Transformer models.
- Enhances obstacle detection accuracy, especially when labeled data is limited.

Conclusion:

GAN-based data augmentation helps create realistic synthetic training images, enabling autonomous vehicle models to learn from a larger and more diverse dataset, resulting in better detection of rare and unexpected obstacles

2. Compare the effectiveness of CNNs versus Vision Transformers for detecting small or sparse objects in high-resolution images.

Answer:

Feature	CNN	Vision Transformer (ViT)
Feature Extraction	Local features	Global features
Training Data	Moderate	Very large
Small Object Detection	Good	Better with large datasets
Computational Cost	Lower	Higher
Speed	Faster	Slower
Accuracy	High	Very High on large datasets
Best Use	Real-time systems	High-resolution image analysis

Conclusion:

- CNN is preferred for real-time embedded vehicle systems.
- Vision Transformers achieve higher accuracy when large datasets and powerful hardware are available.

3. Suggest a real-time inference pipeline for obstacle detection, considering computational constraints in embedded systems.

Answer:

Real-Time Inference Pipeline

1. Camera Image Capture
2. Image Preprocessing
 - Resize
 - Normalize
3. Data Augmentation (during training)

4. Feature Extraction using Lightweight CNN (MobileNet or YOLO)
5. Obstacle Detection
6. Confidence Score Calculation
7. Non-Maximum Suppression (NMS)
8. Obstacle Classification
9. Vehicle Control Decision
10. Braking or Steering Action

Optimization Techniques

- Model Quantization
- Model Pruning
- TensorRT/OpenVINO Optimization
- GPU/NPU Acceleration
- Batch Size = 1 for low latency

These optimizations reduce computation time and enable real-time obstacle detection on embedded automotive systems.